Bias and representativeness in digitized newspaper collections: Introducing the environmental scan

Kaspar Beelen
The Alan Turing Institute, UK

Jon Lawrence
Department of History, University of Exeter, UK

Daniel C.S. Wilson
The Alan Turing Institute, UK

David Beavan
The Alan Turing Institute, UK

Abstract

This article introduces the 'Environmental Scan' as answer to the question of hidden biases in digital heritage collections. Its substantive focus is digitized nineteenth-century British provincial newspapers, and in particular the JISC corpus, a popular, publicly-funded resource for scholars. While multiple papers have meticulously investigated the genesis of such newspaper collections, in the process highlighting the often unacknowledged politics of collection, preservation and dissemination via microfilm and now digitization, our aim is to explore questions of representativeness and bias in new ways by enriching computational analysis of digital corpora with the historical insights that can be derived from a contemporaneous reference source: namely, the Victorian newspaper press directories.

Correspondence:
Kaspar Beelen, The Alan Turing Institute, UK.
E-mail: kbeelen@turing.ac.uk

1 Introduction

As a discipline, History has only hesitantly embraced the use of digital data and computational methods. Historians nonetheless rely increasingly on digitized collections for search and exploration. Studies that pursue research at scale—by harnessing immense datasets for the purposes of historical inquiry—are, however, still comparatively few in number. Data-driven historical research continues to stir debate, partly because the reliance on quantitative, statistical analysis does not sit well with hermeneutics and the careful, highly contextualized interpretation of sources that traditionally prevail in historical scholarship. Especially when applied to datasets whose composition and scope remain ill-understood, data mining can produce insights that lack historical anchoring or context. The controversy on data-driven history is partly attributable to the fact that the most visible articles are published outside traditional history journals—appearing, instead, in high-impact journals such as *Science* or *PNAS*—and that the main authors often are not historians. Encouraged by the size of their data, these publications tend to make sweeping claims, portraying patterns extracted from the data as fundamental cultural and societal

Digital Scholarship in the Humanities, Vol. 38. No. 1, 2023. ©The Author(s) 2022. Published by Oxford University Press on behalf of EADH.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.
https://doi.org/10.1093/llc/fqac037 Advance Access published on 14 July 2022

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

evolutions. Michel *et al.*, for example, coined the term Culturomics, which exploited the immense Google Books corpus to gauge cultural trends, relying mainly on quantitative lexical analysis (Michel *et al.*, 2011). However, despite the size of the dataset, it was still sampled from a relatively narrow set of institutions, primarily university libraries, a strategy which had significant implications for the composition and dynamics of the corpus (Pechenick *et al.*, 2015).[1] Newspaper corpora—the focus of this paper—have emerged as an alternative data source for tracking cultural and even psychological phenomena over time. Newspaper corpora have been used to map cultural transitions, such as technological changes or shifting perceptions of gender relations (Lansdall-Welfare *et al.*, 2017), or used alongside book corpora to plot historical well-being over more than two centuries (Hills *et al.*, 2019). The latter article led to a front-page headline in *The Times* declaring 'Britons were happier when Victoria was on the throne', which further explained that: 'Researchers analysed ... language used in millions of books and newspaper articles and found that the 1880s ... was the country's happiest decade.'[2] Such statements conflate the reading public with the whole population at a time when publishing remained heavily skewed towards middle-class tastes. They also assume that publishers sought faithfully to mirror public attitudes and feelings.

Mining newspapers as entry points to historical well-being or happiness leads to a very pertinent question: whose happiness? Whose past is actually captured by digital collections? Whose voices or perspectives are underrepresented or even completely absent? In this paper, we argue that using historical corpora for analysis at scale—but also, more traditionally, for search—requires a new type of source criticism that explicitly addresses representativeness and biases in digital heritage collections. As Pechenick *et al.* argue, we need to 'characterize the dynamics of the corpus before using these data sets to draw broad conclusions about cultural and linguistic evolution' (Pechenick *et al.*, 2015). By implementing a method we call the Environmental Scan, we contextualize digital newspapers by situating these publications in the wider landscape—their historical 'ecology'—not only estimating the total size of the press, but also inspecting the contemporaneous classifications and discourses that surrounded and shaped the Victorian newspaper industry. This approach, we argue, enables historians to engage more critically with vast amounts of information, identifying gaps and biases in digital data, and anchoring their analysis in a firm grasp of spatial and cultural context even when working at scale. The Environmental Scan of digitized British newspapers we propose in this paper leverages contemporaneous contextual data derived from nineteenth-century Press *Directories* (discussed in detail below). The *Directories* situate the sample of digital papers available to historians in the landscape of all published periodicals, using contemporaneous typologies and classifications of the era. While they are not an objective, unproblematic 'ground truth', they do provide contemporary metadata which allows us to study digital heritage in a historically sensitive manner.

Analysis at scale can only be meaningful if we assume the data represents or 'stands for' (Pitkin, 1967) some aspects of the past. Establishing the representativeness of digital (newspaper) collections always remains an approximation. As we argue below, the point of this exercise is not to reduce 'representativeness' or 'bias' to a simple score, but to understand the composition and dynamics of a corpus and incorporate its characteristics into the historical analysis. We demonstrate our approach with a specific case study that probes political and other biases in a corpus of digitized British provincial newspapers. We focus on the JISC corpus (JISC I and JISC II), a subset of British Library Newspapers originally digitized by the UK's Joint Information System Committee in partnership with the British Library. We had access to the corpus through a collaboration agreement with the British Library, but researchers may also acquire the data under licence from Gale Digital. We have cleaned and de-duplicated the JISC metadata and will redeposit this enhanced version with BL Labs for use by future scholars.[3] Another reason to scrutinize the JISC corpus was to assess how successfully its creators had realized their aim of curating a balanced sample of the so-called 'provincial' press. Traditionally, newspaper research in Britain relied heavily on metropolitan papers, and especially the well-indexed *Times*, which is regularly cited as the authoritative voice on almost any issue. But, as recent studies in media history emphasize (Hobbs, 2018, 2013), historians' preoccupation with London-based newspapers conflicts

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

with the preferences of the Victorian reader, who generally preferred a local newspaper. The provincial press had an immense influence on Victorian culture but remains understudied. This paper, therefore, aims to provide future scholars with a better understanding of the scope and limits of the digitized provincial newspapers they are working with.

We demonstrate how using newly-created metadata derived from Victorian reference works (annual newspaper press directories) enables historians to evaluate the representativeness of digital newspaper collections by quantifying which types of newspapers are over- or under-represented. We find that the JISC corpus contains a conservative bias and fails to capture a significant category of 'neutral' and 'independent' papers—the former largely correlating with local newspapers—leading to an increasing divergence between sample and population as the nineteenth century progresses. We also explore representativeness in relation to price, place of publication and types of newspaper (as derived from the Press *Directories*' short newspaper 'portraits'). On price, we find that the JISC corpus under-samples cheaper newspapers aimed at a mass readership, especially for the 1860s and 1870s, when both press and polity were taking a decisive leap towards democracy. On place, we show that JISC's curators succeeded in their aim to offer broad nationwide coverage, although with the caveat that the Southeast of England and some individual large counties are underrepresented. Our discursive analysis of the Press Directories largely confirms the bias towards political over non-political titles, and demonstrates that newspapers that foregrounded locality and quotidian interests tended to be under-represented.

Last, we emphasize that the conclusions of this paper should not be interpreted as a criticism of the JISC corpus. The purpose of the Environmental Scan is not to designate a corpus as 'good' or 'bad', but to give historians instruments to assess how digitization affects the outcomes of their analysis and how appropriate the data are for their research questions. It can also be a powerful tool for those making future decisions about digitization. As we argue in the conclusion, bias is not a ground for rejecting a corpus, but nor should historians and curators simply ignore issues of representativeness. Ultimately, both analogue and digital information require an appropriate form of source criticism, and studying bias in digital collections needs to be an intrinsic part of any research that uses these materials.

2 The Environmental Scan, or, How to Read Digitized Newspapers in Context

The Environmental Scan facilitates source criticism at scale by contextualizing the information present in a digitized collection. The scan moves in two directions: deep and wide. The wide scan studies islands of digitized data in relation to the 'landscape', with the latter serving as a metaphor for the totality of information, in our case about the Victorian press. The wide scan is concerned with measuring the representativeness of digitized sources: how do the characteristics of the digital sample compare with those of the wider known landscape? And, insofar as it is not representative, where do the major biases of the sample lie? The deep scan uncovers the latent or hidden dimensions that structure a dataset. 'Time' is an important example of such a dimension. Digital newspapers come with minimal metadata that includes time stamps, but other aspects are often not captured such as information about both production and consumption: who produced papers and who was likely to read them? Digitization has given historians access to a lot of materials, but has often situated it in a contextual vacuum. The deep scan attempts to automatically generate metadata from contextual resources—for all known newspapers, not only those that are digitized—and thereby provides much richer profiles for each data point in a given collection. The deep scan encodes context; adding layers of metadata to articulate the social, political, and historical dimension of the (digitized) press.

This paper embraces mass digitization initiatives: it is not advocating a return to 'small' but very well-described data. However, we do argue that even large datasets require context in order to be usable for scholarly research. Current collections—such as the largest and most routinely used ones such as Google *n*-grams and the British Newspaper Archive—would be more valuable to scholars if their content was well described, both in the sense of the deep scan (richer metadata) and the wide scan (information about

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

content selection and biases in the composition). We apply these insights of the Environmental Scan to digital newspapers, showing how the ambitions of source criticism at scale can be realized practically, and how this changes critical approaches to digital collections.

We do not propose our method as a universal procedure for source criticism at scale, or a roadmap that other scholars can simply copy and execute. As we demonstrate below, the Environmental Scan of newspapers draws upon a specific set of historical *Directories* that depict a detailed and (almost) complete picture of the Victorian press landscape. Similar resources may not be available for other collections and periods. However, some principles outlined here could be applied to other collections: digitized book collections, for example, can be linked to and compared with more complete catalogues, which also often provide metadata. Ultimately, source criticism—whether conducted traditionally, or at scale—depends on the collection under scrutiny and the resources available. Nonetheless, the paper aims to encourage research on data bias, discussing general principles and demonstrating their practical application. In so doing, we hope to spark a wider debate within the (Digital) Humanities, to ensure that collections better meet the requirements of scholarly, historical research.

3 Literature Review

3.1 Newspaper collections as research objects

Digital collections are increasingly interrogated and explored for historical research. They have facilitated access and, as a result, become a critical platform where scholars find and select sources. The reliance on such digitized resources can be problematic, since, oftentimes, the underlying data remain opaque. As Yale points out, archives (both digital and analogue) are not just 'sources of history . . . they are also its subjects, sites with histories and politics of their own' (Yale, 2015). These observations apply all the more to newspaper collections. With digitization and full-text indexing, millions of pages are just a few keystrokes (or lines of code) away (Milligan, 2013). However, the manner in which digitization proceeds, while documented, is nonetheless hybrid, and even though scholars rely on digital newspapers for their research, they are confronted with an opaque pool of information whose history and content remains poorly understood. Answers to critical questions in this regard—'why material was chosen and curated, how it was obtained, and from which specific source material it was transformed into a digital copy'—are hard to come by (Hauswedell *et al.*, 2020). But these issues are critical, because faced with such a wealth of historical newspapers in global libraries, digitization initiatives are necessarily highly selective. It is estimated that at present only about 6% of the British Library's collection of 750 million pages of British newspapers has been digitized, and that the JISC corpus, the most freely available digital corpus, represents less than 1% (approximately five million pages) (Fyfe, 2016; King, 2005; Shaw, 2009, 2006).

Many factors influence the selection of titles for digitization, including prior decisions about microfilm preservation (it is much more expensive to digitize from hard copy than from well-preserved microfilm), contemporary research interests, perceived quality, and copyright considerations (Fyfe, 2016; Hauswedell *et al.*, 2020). Both the creation of a digital collection and the ramifications that result from the selection process remain little understood, but recent work has helped to elucidate the hybrid process of preservation and digitization. Conducting semi-structured interviews with librarians, archivists and digital content managers has clarified the decisions about inclusion and exclusion that underpin digital newspaper collections, creating a 'highly detailed snapshot of current digital newspaper practices' (Hauswedell *et al.*, 2020). Fyfe has taken a long-term view of historic newspaper collections, noting how the analysis of these materials focuses either on the Victorian past or the digital present, overlooking the interval in between (Fyfe, 2016). Uncovering the 'twentieth-century transmission histories', the complex process of data survival and transformation, is crucial for understanding the shape of digitized collections. The 'archaeology of data' championed by Fyfe emphasizes how the material and media histories of newspaper collecting have left their imprint on today's digital resources (see also Mak, 2014, Bode 2017 and Cordell, 2017). Our analysis builds on these histories to develop an understanding of newspaper

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

collections as digital samples drawn from a wider universe of information. We aim to explore how the shape of these corpora affects research outcomes by mapping and quantifying the effect of past choices on the composition of our scholarly resources.

Historical newspapers often serve as the example par excellence of 'big data' in the humanities, and historians have lauded as well as criticized the increasing use of such digital resources. Such reflections have been less concerned with the properties of the data than with its availability (or lack thereof, for example, the interfaces and paywalls that mediate and structure access (Brake, 2012)). But understanding missing data and gaps (what's in and what's out) is critical even for large collections and it requires new methods and strategies. Discussing the digitized British Library Newspapers, Brake asks 'So what *do* we have?' (emphasis in the original) reminding historians that a large majority of newspapers are still absent, which makes the digitized materials only 'indicative, icebergs on a surface that constantly remind readers of the invisible dangers, and riches, in their immediate environs' (Brake, 2012).

Of course, the problem of selective digitization goes beyond newspaper collections. Biber offers a more linguistic perspective focussing on corpus construction and emphasizing the intricate relation between the theoretical framework, sampling strategies and corpus design (Biber, 1993). Discussing the digitized book corpus Eighteenth Century Collections Online (ECCO), Holahan notes that researchers tend to '[dip] into this vast, opaque pool for material without understanding what, exactly, [they] are searching within' (Holahan, 2021). Using probabilistic analysis, Lahti *et al.* (2020) report a bias towards female authors, pamphlets, and recent publications in ECCO. In a similar vein, Algee-Hewitt *et al.* (2016) distinguish between the 'published', the 'archive' and the 'corpus' which they portray as 'three Russian dolls', or matryoshkas, 'fitting neatly into one another'. The archive contains the set of preserved newspapers, from which a corpus is derived as a bundle of materials selected for specific research purposes. The authors argue that, at least for literary studies, the three layers are converging, with the corpus containing almost all published works, allowing a 'total history of literature'. While, in theory, the same observations could apply to British newspaper corpora, the reality is different. First, even though digitization is gaining pace, as noted, the digitized collection remains a fraction of library holdings, which in turn represent only a subset of all published newspapers. Secondly, in countries such as Britain where digitization has been arranged on a commercial basis, only a very small fraction of the collection is freely accessible for scholarly research at scale: a trend that present funding arrangements seem unlikely to change (Brake, 2012; Fyfe, 2018).[4]

3.2 Bias and representativeness

In seeking to understand the relationship of a sample to the wider population, the Environmental Scan necessarily involves the 'foundational yet slippery concept' of 'representativeness' (Chasalow and Levy, 2021). When discussing the kind of sampling involved in the creation of a historical newspaper collection such as the JISC corpus, notions of representativeness require us to engage with the statistical ('is the digital sample representative of the press?') as well as the political interpretation of the term ('who is represented by the data?'). These interpretations are not necessarily aligned and can lead to contradictory assessments of bias (understood as a divergence from representativeness). When thinking about the representativeness of a digital newspaper collection, our reference point is the totality of information published, i.e. the newspaper landscape we know about and we perceive representation bias as the divergence (with respect to a set of known variables) between the digitized press and the known newspaper landscape. Bias, then, measures the extent to which a sample proportionally diverges from the larger universe of information.

But framing the issue of bias as internal to the press risks circumventing fundamental questions about big data as a lens on the past. Rather than treating bias as a property of our data that can be removed, we need to recognize it as intrinsic to the historical materials we work with; something to be understood and articulated, rather than 'solved'. Bias exists prior to any sampling: even if all newspapers were digitized, such a corpus would still reflect the hierarchies of power that governed Victorian society; reproducing its social prejudices and stereotypes, (Garg *et al.*, 2018) rather than reflecting the diversity of opinions and perspectives of the whole population. Hence the

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

problem with claims to measure aggregate happiness or well-being (Hills *et al.*, 2019). Unbiased data—even as an idea—is essentially ahistorical data. Historical bias, then, comprises the footprint of social and historical inequalities left on our sources, as opposed to representation bias which concerns the divergence between sample and population. 'Unbiasing' becomes an inherently normative choice between equal representation—the 'equal representation of each distinct subgroup of the user population' (Chasalow and Levy, 2021)—or proportionate representation (the avoidance of under- or over-representation according to a set of variables). Both notions can be in direct conflict with each other: creating a proportionally balanced set may hurt the inclusion of already underrepresented voices (and thereby increase historical bias). In the case of Victorian newspapers, proportionately representing the sizable liberal press could impinge on the presence of minority radical or working-class periodicals. The goal of the Environmental Scan is not to avoid bias (or accomplish perfect representativeness), but rather to make the issue more visible and give researchers a clearer picture of the gaps and imbalances in their data as well as tools to engage with this issue (Bode, 2020).

3.3 Provincial newspapers and JISC

One impact of newspaper digitization has been to encourage historians to extend their perspective beyond the metropolitan voices which have long dominated scholarship and produced a London-centric view (Hobbs, 2013). The *British Library Newspapers* collection published by Gale Digital has, for example, brought the provincial press within reach; although London was the biggest centre of newspaper production, the Victorian press landscape contained a myriad of thriving and distinctive local cultures (Brake, 2015). The provincial press was a national phenomenon—connected to London but also other regions, cities and towns through a nationwide network—with a 'greater influence on Victorian culture than any other type of print' (Hobbs, 2018).

The Victorian era witnessed a rapid growth in both the number and the circulation of provincial newspaper titles, driven by both technological change and political reforms (Wiener, 2009, 1969). In 1856, there were 370 provincial English newspapers—already twice as many as the 152 London papers. But by 1860, the number of provincial titles had leapt to 743 and by 1880 it was over 1,000 (Lee, 1976; Table 28). The growth in daily provincial titles was even more dramatic. According to Lee, there were only two such titles before the Repeal of the Stamp Duty in 1855, by 1868 there were forty-three, and at the close of the century this had reached a peak of 171 ((Lee, 1976), Tables 1 and 24). Compared to 1850, the provincial press had grown six-fold by the end of the century, while the London press only doubled (Hobbs, 2018). Hobbs also argues that provincial newspapers commanded a wider readership: 'For most of the population, provincial newspapers were not on the fringes of nineteenth-century print culture, but at the very heart of it'. The JISC digitized newspaper collection is the most freely available corpus of nineteenth-century British provincial newspapers and was created in two tranches in the 2000s in a collaboration between the British Library, Gale Cengage Learning and the Joint Information Systems Committee, now Jisc, the UK's higher and further education infrastructure body. Formally known as the British Newspapers 1800–1900 project, this initiative sought to digitize a range of important national and regional newspaper titles to provide 'countrywide coverage with the breadth and depth to form a virtual key to many other provincial newspapers of the same date' (Fyfe, 2016, p. 562 citing King, 2008). The aim was to produce something akin to a virtual version of *Palmer's Index* to *The Times* for the British provincial press.[5] Initially, forty-eight titles were selected to offer, as near as possible, nationwide coverage. There was a strong preference for titles deemed to approximate most closely to regional 'papers of record' (most like the nineteenth-century *Times*). The selection was further slanted in favour of titles with long, continuous holdings at the British Library and with no present-day copyright claimants. The second tranche followed similar lines, but placed a stronger emphasis on local titles and extended digitization back to the eighteenth century wherever possible (Fyfe, 2016, pp. 561–567).

The JISC corpus comprises about five million digitized newspaper pages, which sounds impressive—definitely too much to read in a lifetime (a criterion often adduced to distinguish 'big' from 'small' data in the Humanities). For comparison, the largest British digitized newspaper corpus, the result of the current

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

collaboration between the British Library and genealogy company Find My Past, contained ca. 45 million pages in September 2021,[6] which sounds even more impressive, but still constitutes only about 6% of the estimated total of 750 million newspaper pages in the British Library's collection. In this paper, we focus on the provincial (i.e. non-Metropolitan) newspapers of the JISC corpus published after 1846, which is the first year for which we have historical information on the Victorian newspaper landscape derived from press directories.[7]

3.4 Newspaper press directories

To establish the magnitude of the Victorian press has always been a challenge: contemporaries themselves struggled to fathom the scope of this complex and expanding industry. In 1846, Charles Mitchell published the first edition of his *Newspaper Press Directories*, a new type of publication that aimed to provide a 'dignified and permanent' record of the British press primarily as a guide to would-be advertisers (Gliserman, 1969). Other editions followed, initially at irregular intervals (Brake, 2015). From 1858 the *Directories* appeared annually and Mitchell's attempt to codify the press was recognized by the Post Office in 1861, which considered it as an authoritative list of London and provincial newspapers (Gliserman, 1969). Its status as a respectable and reliable resource persisted into the twentieth century, some even termed it the 'Whitaker of the press' (O'Malley, 2015). *Directories*, as a genre, were symptomatic of wider cultural and philosophical trends in the nineteenth century. Following the traditions of the Encyclopaedists and Bentham, various schools of thinkers placed the systematic collection, classification and tabulation of information, at the heart of rational and scientific inquiry. Not only nature but also society, industry, and governance were increasingly understood through 'statistics'. The foundation of the Manchester Statistical Society in 1833, bears testimony to this phenomenon, as did the expansion of the State in the same period: policymaking—especially in the social and economic sphere—required governments to have a stronger numerical grip on their populations. The detailed and nationwide sources historians can consult today, such as the Census and Ordnance Survey maps, are manifestations of this urge to count, classify, and study society (Poovey, 1995; O'Malley, 2015).

Other reference works such as Kelly's *Post Office Directories*, Bradshaw's *Railway Timetables*, Palmer's *Index to The Times*, presented systematic collections of statistical information to a newly literate and numerate public (Vincent, 2014; Brake, 2015). In addition to this context, Mitchell's attempt to chart a proliferating press was embedded in trade and commerce. The primary function of the *Directories* was to promote the interests of proprietors and serve as a guide to advertisers. In this sense Mitchell was himself a crucial node in the network of the Victorian press, mediating between journalists, printers, and the advertising industry.

While historians have frequently consulted press directories, this has often been to collect 'facts' on newspapers and so these texts have rarely received historical investigation in their own right. But, as Laurel Brake, observes: 'Annual listings in the press directories are of immense value to scholars: not only do they provide a trajectory of prices hard to find elsewhere except through examinations of issues, but they also represent the industry diachronically and synchronically, offering information about changes in titles, readerships, publishers, illustrations, and geographical distribution' (Brake, 2015). Of course, even Mitchell cannot be viewed as offering a ground truth of the nineteenth-century newspaper landscape, but given its authority and longevity, the *Directories* represent a valuable and under-studied resource on the press. To the best of our knowledge, this paper is the first to analyse historical directories using computational methods. The few papers that treat press directories as a historical resource focus on describing the genre (Brake, 2015; Gliserman, 1969) or understanding the contemporaneous classifications and understandings of newspapers, such as the emergent (but still ill-defined) category of 'national newspaper' before 1930 (O'Malley, 2015).

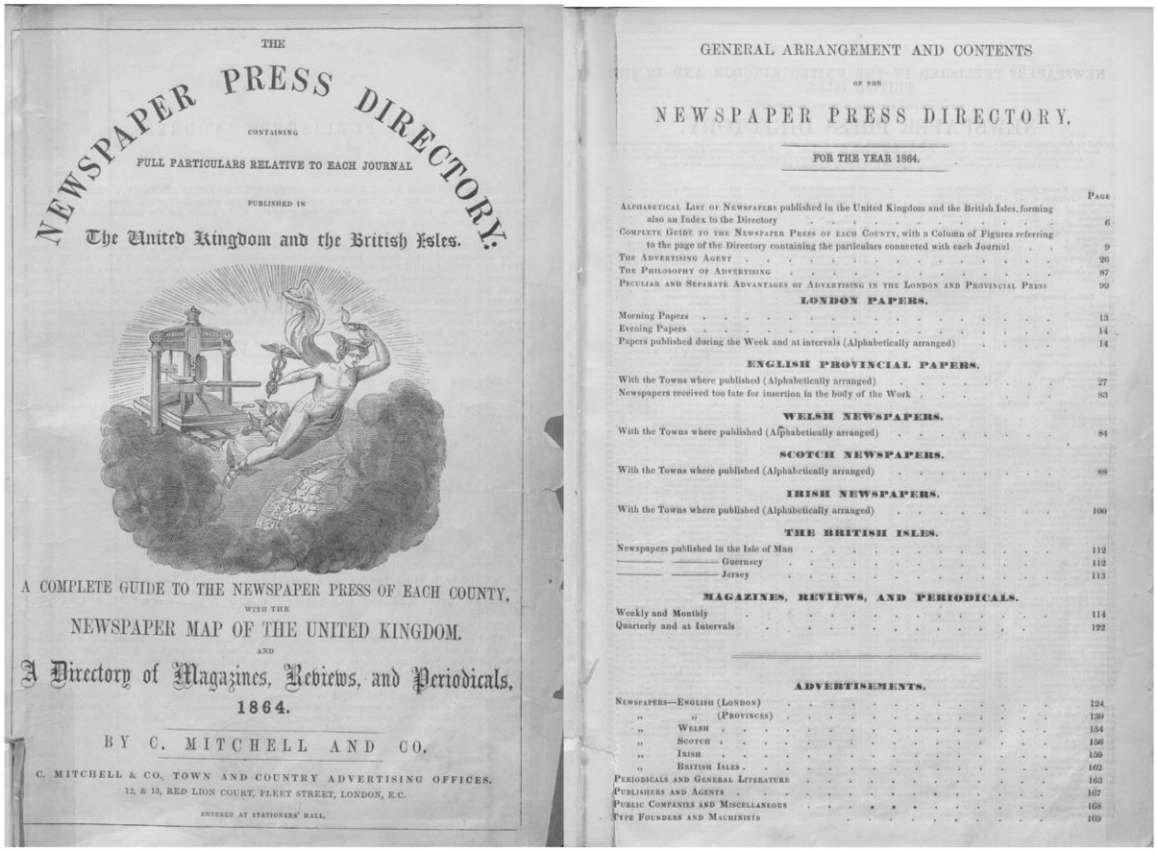
What information did Mitchell's *Newspaper Press Directories* contain? First and foremost, they provided a geographical index, with papers ordered along an implicit value hierarchy determined by place and frequency. The London *Dailies* stand at the top of this hierarchy, followed by the weekly, monthly and even quarterly metropolitan periodicals (Brake, 2015, see also Fig. 1). Then came the Provincial Press (the

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

papers printed in England and Wales, outside of London), followed by the Irish and Scottish Press, and finally the Crown dependencies. But the geography of the Directories later stretched beyond these Isles, as Mitchell 'positioned the industry as part of the British Empire' by including papers from the USA and Canada, and from 1886 an 'Indian and Colonial Supplement' (O'Malley, 2015).

The emphasis on place is apparent from the structure of the directories: newspapers are ordered alphabetically by district (Fig. 2 left) and often contain very detailed descriptions of places where they circulated (Fig. 2 right). For the Cambridge Chronicle, shown in Fig. 2, the entry lists up to sixteen districts, towns, parishes, and other places where it circulated. The entry also captures political leaning ('Conservative'), price ('3 and 4 pennies'),[8] periodicity (Saturdays) and 'seniority', or the date the paper was established ('May 1748'). After the extensive list of place names, follows a 'portrait': a summary of the newspaper's emphasis, scope, interests, and audiences. At the end come the names of proprietors, printers, and/or publishers. The extensive attention given to provincial newspapers (the detailed descriptions and place enumerations) challenges the supposed primacy of the Metropolitan papers (Brake, 2015), and supports the current historiographical emphasis on local newspapers as a preeminent source about Victorian culture.

Lastly, Mitchell's notion of 'newspapers' itself remains ambiguous. Though set apart from periodicals—which only appear from 1860 onwards, and even then, are more summarily discussed—the distinction was fluid and open to change. Publications



THE NEWSPAPER PRESS DIRECTORY.

CONTAINING FULL PARTICULARS RELATIVE TO EACH JOURNAL PUBLISHED IN The United Kingdom and the British Isles.

A COMPLETE GUIDE TO THE NEWSPAPER PRESS OF EACH COUNTY, WITH THE NEWSPAPER MAP OF THE UNITED KINGDOM, AND A Directory of Magazines, Reviews, and Periodicals, 1864.

BY C. MITCHELL AND CO.

C. MITCHELL & CO., TOWN AND COUNTRY ADVERTISING OFFICES, 12, & 13, RED LION COURT, FLEET STREET, LONDON, E.C.

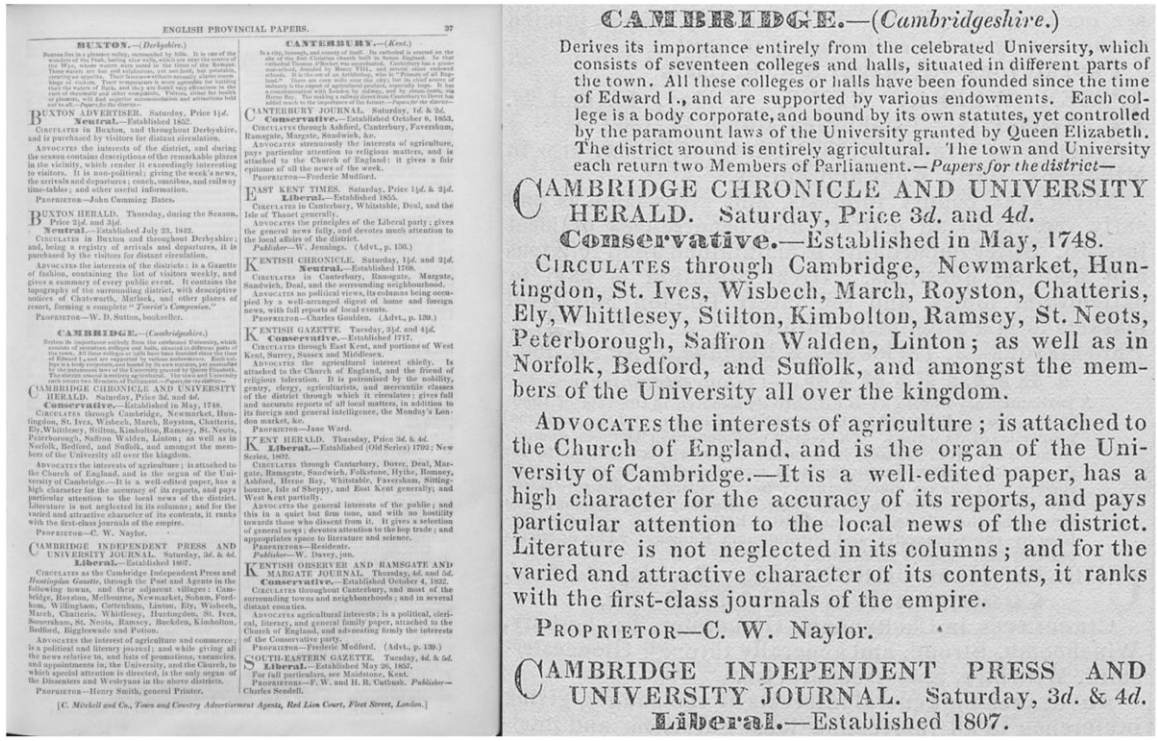
ENTERED AT STATIONERS' HALL.

GENERAL ARRANGEMENT AND CONTENTS OF THE NEWSPAPER PRESS DIRECTORY, FOR THE YEAR 1864.

	Page
Alphabetical List of Newspapers published in the United Kingdom and the British Isles, forming also an Index to the Directory	6
Complete Guide to the Newspaper Press of each County, with a Column of Figures referring to the page of the Directory containing the particulars connected with each Journal	9
The Advertising Agent	26
The Philosophy of Advertising	87
Peculiar and Separate Advantages of Advertising in the London and Provincial Press	99
LONDON PAPERS.	
Morning Papers	13
Evening Papers	14
Papers published during the Week and at intervals (Alphabetically arranged)	14
ENGLISH PROVINCIAL PAPERS.	
With the Towns where published (Alphabetically arranged)	27
Newspapers received too late for insertion in the body of the Work	83
WELSH NEWSPAPERS.	
With the Towns where published (Alphabetically arranged)	84
SCOTCH NEWSPAPERS.	
With the Towns where published (Alphabetically arranged)	88
IRISH NEWSPAPERS.	
With the Towns where published (Alphabetically arranged)	100
THE BRITISH ISLES.	
Newspapers published in the Isle of Man	112
——— ——— Guernsey	112
——— ——— Jersey	113
MAGAZINES, REVIEWS, AND PERIODICALS.	
Weekly and Monthly	114
Quarterly and at Intervals	122
ADVERTISEMENTS.	
Newspapers—English (London)	124
,, ,, (Provinces)	139
,, Welsh	154
,, Scotch	156
,, Irish	159
,, British Isles	162
Periodicals and General Literature	163
Publishers and Agents	167
Public Companies and Miscellaneous	168
Type Founders and Machinists	169

Fig. 1 Pages taken from Mitchell's 1864 Newspaper Press Directory. The image on the left shows the front page, the one on right the table of contents (BL Gen. Ref. Coll. Mic.A.3594-3606)

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025



ENGLISH PROVINCIAL PAPERS. 37

BUXTON.—(*Derbyshire.*)

BUXTON ADVERTISER. Saturday. Price 1½d. Neutral.—Established 1852.
CIRCULATES in Buxton, and throughout Derbyshire, and is purchased by visitors for distant circulation.
ADVOCATES the interests of the district, and during the season contains descriptions of the remarkable places in the vicinity, which render it exceedingly interesting to visitors. It is non-political; giving the week's news, the arrivals and departures; coach, omnibus, and railway time-tables; and other useful information.
PROPRIETOR—John Cumming Bates.

BUXTON HERALD. Thursday, during the Season. Price 1d. and 3d. Neutral.—Established July 23, 1842.
CIRCULATES in Buxton and throughout Derbyshire; and, being a registry of arrivals and departures, it is purchased by the visitors for distant circulation.
ADVOCATES the interests of the district; is a Gazette of fashion, containing the list of visitors weekly, and gives a summary of every public event. It contains the topography of the surrounding district, with descriptive notices of Chatsworth, Matlock, and other places of resort, forming a complete "*Tourist's Companion.*"
PROPRIETOR—W. D. Sutton, bookseller.

CAMBRIDGE.—(*Cambridgeshire.*)
Derives its importance entirely from the celebrated University, which consists of seventeen colleges and halls, situated in different parts of the town. All these colleges or halls have been founded since the time of Edward I., and are supported by various endowments. Each college is a body corporate, and bound by its own statutes, yet controlled by the paramount laws of the University granted by Queen Elizabeth. The district around is entirely agricultural. The town and University each return two Members of Parliament.—*Papers for the district—*
CAMBRIDGE CHRONICLE AND UNIVERSITY HERALD. Saturday, Price 3d. and 4d. Conservative.—Established 1748.
CIRCULATES through Cambridge, Newmarket, Huntingdon, St. Ives, Wisbech, March, Royston, Chatteris, Ely, Whittlesey, Stilton, Kimbolton, Ramsey, St. Neots, Peterborough, Saffron Walden, Linton; as well as in Norfolk, Bedford, and Suffolk, and amongst the members of the University all over the kingdom.
ADVOCATES the interests of agriculture; is attached to the Church of England, and is the organ of the University of Cambridge.—It is a well-edited paper, has a high character for the accuracy of its reports, and pays particular attention to the local news of the district. Literature is not neglected in its columns; and for the varied and attractive character of its contents, it ranks with the first-class journals of the empire.
PROPRIETOR—C. W. Naylor.

CAMBRIDGE INDEPENDENT PRESS AND UNIVERSITY JOURNAL. Saturday, 3d. & 4d. Liberal.—Established 1807.
CIRCULATES as the Cambridge Independent Press and *Huntingdon Gazette*, through the Post and Agents in the following towns, and their adjacent villages: Cambridge, Royston, Melbourne, Newmarket, Soham, Fordham, Willingham, Cottenham, Linton, Ely, Wisbech, March, Chatteris, Whittlesey, Huntingdon, St. Ives, Somersham, St. Neots, Ramsey, Buckden, Kimbolton, Bedford, Biggleswade and Potton.
ADVOCATES the interest of agriculture and commerce; is a political and literary journal; and while giving all the news relative to, and lists of promotions, vacancies, and appointments in, the University, and the Church, to which special attention is directed, is the only organ of the Dissenters and Wesleyans in the above districts.
PROPRIETOR—Henry Smith, general Printer.

[*C. Mitchell and Co., Town and Country Advertisement Agents, Red Lion Court, Fleet Street, London.*]

CANTERBURY.—(*Kent.*)

CANTERBURY JOURNAL. Saturday, 1d. & 2d. Conservative.—Established October 8, 1853.
CIRCULATES through Ashford, Canterbury, Faversham, Ramsgate, Margate, Sandwich, &c.
ADVOCATES strenuously the interests of agriculture, pays particular attention to religious matters, and is attached to the Church of England: it gives a fair epitome of all the news of the week.
PROPRIETOR—Frederic Mudford.

EAST KENT TIMES. Saturday. Price 1½d. & 3½d. Liberal.—Established 1855.
CIRCULATES in Canterbury, Whitstable, Deal, and the Isle of Thanet generally.
ADVOCATES the principles of the Liberal party; gives the general news fully, and devotes much attention to the local affairs of the district.
Publisher—W. Jennings. (Advt., p. 136.)

KENTISH CHRONICLE. Saturday, 1½d. and 3½d. Neutral.—Established 1768.
CIRCULATES in Canterbury, Ramsgate, Margate, Sandwich, Deal, and the surrounding neighbourhood.
ADVOCATES no political views, its columns being occupied by a well-arranged digest of home and foreign news, with full reports of local events.
PROPRIETOR—Charles Goulden. (Advt., p. 139.)

KENTISH GAZETTE. Tuesday, 3½d. and 4½d. Conservative.—Established 1717.
CIRCULATES through East Kent, and portions of West Kent, Surrey, Sussex and Middlesex.
ADVOCATES the agricultural interest chiefly. Is attached to the Church of England, and the friend of religious toleration. It is patronized by the nobility, gentry, clergy, agriculturists, and mercantile classes of the district through which it circulates; gives full and accurate reports of all local matters, in addition to its foreign and general intelligence, the Monday's London market, &c.
PROPRIETOR—Jane Ward.

KENT HERALD. Thursday. Price 3d. & 4d. Liberal.—Established (Old Series) 1792; New Series, 1807.
CIRCULATES through Canterbury, Dover, Deal, Margate, Ramsgate, Sandwich, Folkstone, Hythe, Romney, Ashford, Herne Bay, Whitstable, Faversham, Sittingbourne, Isle of Sheppy, and East Kent generally; and West Kent partially.
ADVOCATES the general interests of the public; and this in a quiet but firm tone, and with no hostility towards those who dissent from it. It gives a selection of general news; devotes attention to the hop trade; and appropriates space to literature and science.
PROPRIETORS—Residents.
Publisher—W. Davey, jun.

KENTISH OBSERVER AND RAMSGATE AND MARGATE JOURNAL. Thursday, 4d. and 5d. Conservative.—Established October 4, 1832.
CIRCULATES throughout Canterbury, and most of the surrounding towns and neighbourhoods; and in several distant counties.
ADVOCATES agricultural interests; is a political, clerical, literary, and general family paper, attached to the Church of England, and advocating firmly the interests of the Conservative party.
PROPRIETOR—Frederic Mudford. (Advt., p. 130.)

SOUTH-EASTERN GAZETTE. Tuesday, 4d. & 5d. Liberal.—Established May 26, 1857.
For full particulars, see Maidstone, Kent.
PROPRIETORS—F. W. and H. R. Cuttland. *Publisher*—Charles Sendell.

Fig. 2 Page taken from the 1863 edition of Mitchell's Newspaper Press Directories showing the 'English Provincial Newspapers' (left). The image at the right-hand side is an excerpt from the same page focussing on Cambridge district and *The Cambridge Chronicle*

were categorized as newspapers based on whether they carried news, their frequency, circulation but also their perceived status. Especially in the case of London Papers, many of the listed titles would today be considered as periodicals (examples are The *Athenaeum*, *Bell's Weekly*, etc.).

4. Sources and Data

This section describes, in more formal and quantitative terms, the data used in this paper. In 2019, the Living with Machines project commissioned the British Library Imaging Studio to scan all editions of Mitchell's Press Directories printed between 1846 and 1920. These were further processed and enriched by Beelen.[9] For this paper, we used a curated subsample of the directories, for which semi-manually corrected the data.[10] Table 1 shows an example of the tabular data extracted from the scans. It only shows the fields used in this paper, a more extensive overview is given in Online Appendix A.

Figure 3 plots the number of newspapers by year we detected in the digitized press directories, differentiating between the London newspapers and those circulating in the English 'provinces', Wales, Scotland, and Ireland. It confirms the proliferation of provincial newspapers during the second part of the nineteenth century, which plateaus around 1900, the terminal date for the JISC corpus (and hence this paper).

To understand the contours of the JISC corpus, we linked the digitized, provincial newspapers in this collection to entries in the directories. We obtained a list of JISC newspaper titles from the British Library[11], which as the black line Fig. 4 shows, remains stable for most of the period, hovering between 35 and 40 titles per year. The grey line in Fig. 4 shows the number of linked titles for each year, which reveals a small discrepancy, but one that is stable for the whole period

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

Table 1. Example output of the processing pipeline for the year 1883

Title	Politics	District	County	Price	Date	Description	Link_id
Altrincham and Bowdon guardian	Neutral	Bowdon	Cheshire	2d; 3d	1862	Takes no part in politics, [. . .].	null
Brackley observer	Liberal	Brackley	Buckinghamshire	1½d	1858	Taking a liberal view of politics [. . .].	CID_00570

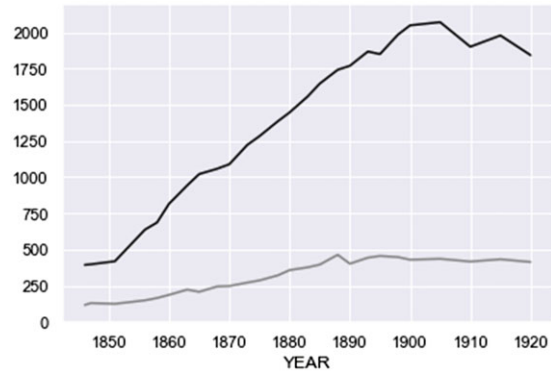


Fig. 3 Number of newspaper titles by year: provincial (black) and London (grey) newspapers

and unlikely to affect the results that we report below.[12] Linking back to the notion of representativeness: the stable corpus size (expressed in the number of titles) does not reflect the actual proliferation of the press in the same period. Over time, the digital sample makes up a smaller percentage of the total landscape of known papers. And the decline is rather drastic, from 10% around 1850 to less than 2 per in 1900. While this evolution is perhaps unsurprising, it is nonetheless something easily overlooked when conducting digital searches or analysis at scale. Of course, the divergence in Fig. 4 does not necessarily imply that the JISC corpus becomes more unrepresentative over time, it does mean, however, that the risk of deviation between sample and population increases. The Environmental Scan below will look in more detail at how this affects the presence of bias in the JISC corpus.

5. Analysis: Bias in Digitized Newspapers

Whose past is captured by a digitized (newspaper) collection? Who does the data represent? By contextualizing information in a collection, the Environmental Scan articulates the biases and gaps it contains: what's in and what's out? Such contextualization makes the inherent (but often latent) polyphony of the data explicit—which voices are present, which are absent—and allows scholars to have a clearer view into an otherwise opaque pool of texts.

5.1 Political bias in the JISC corpus

To quantify political bias in the JISC collection we apply the contemporaneous political typology created by Mitchell for the provincial press, which was fine-grained, containing around thirty different head labels—such as 'liberal' and 'conservative'[13]—that also appeared in hybrid forms (e.g. 'liberal-conservative'), extending the classification scheme to a total of approximately seventy items. Some distinctions appear obscure, for example, besides 'neutral', Mitchell deploys 'no-politics', 'non-party', and 'non-political' as descriptors. Moreover, the composite labels appear in several forms, e.g. separated by a space ('Liberal Conservative'), hyphen ('Liberal-Conservative'), or in a conjunction ('Liberal & Conservative'). In the analysis below, we try to stay close to these actors' categories, but also simplify the schema by grouping semantically similar classes in the long tail of the frequency distribution.[14]

Figure 5 shows the distribution of labels for the original (left) and simplified typology (right). The distributions are highly skewed in both cases with a handful of labels appearing very often, followed by a tail of very infrequently occurring tokens. By far the most common classifications are 'liberal' and 'conservative', followed by 'neutral'/'non-aligned' and 'independent'. While 'independent' and 'neutral' seem semantically close, only the former co-occurs with 'liberal' and 'conservative' in composite labels, suggesting it carries political connotations, but only loosely adhered to a particular party (Beales, 1967).

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

The *Directories*, as a political survey of the press industry, register shifts in the political orientation of newspapers over time; and so, representativeness is not a stable attribute of the data; rather, a moving feast, sensitive to transformations of the sample and newspaper landscape. In this section, we look at the proportional likeness between sample and population and compute a 'bias' measure based on information theory that captures fluctuations on both levels in one score. Figure 6 plots the presence—in proportional terms—of political voices in both Mitchell's and the JISC corpus. The orange lines, for example, express the ratio of liberal titles (i.e. the number of liberal papers divided by all the papers) for both the JISC

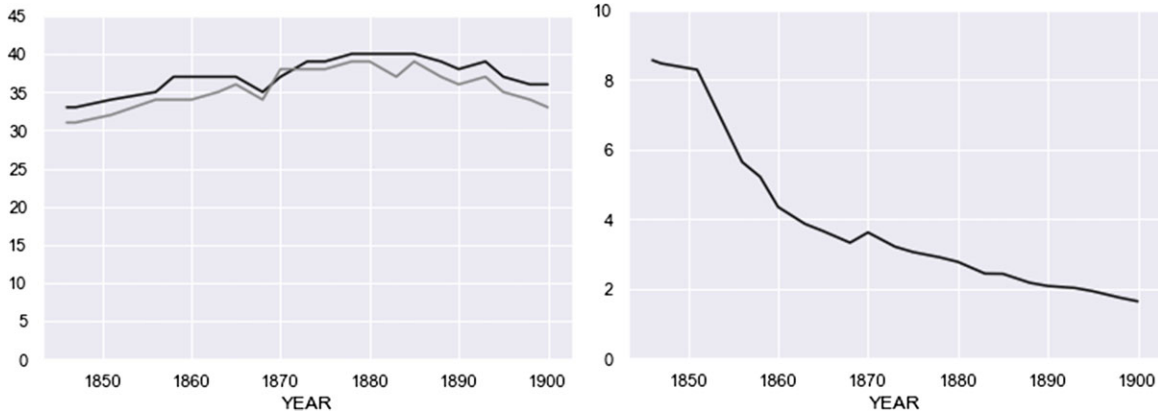


Fig. 4 Comparing the number of JISC provincial newspapers over time (black); grey counts the number of links between the digitized Press Directories and JISC metadata by year (left). The figure on the right plots the JISC provincial newspaper titles as a percentage of all provincial titles in Mitchell's Directories.

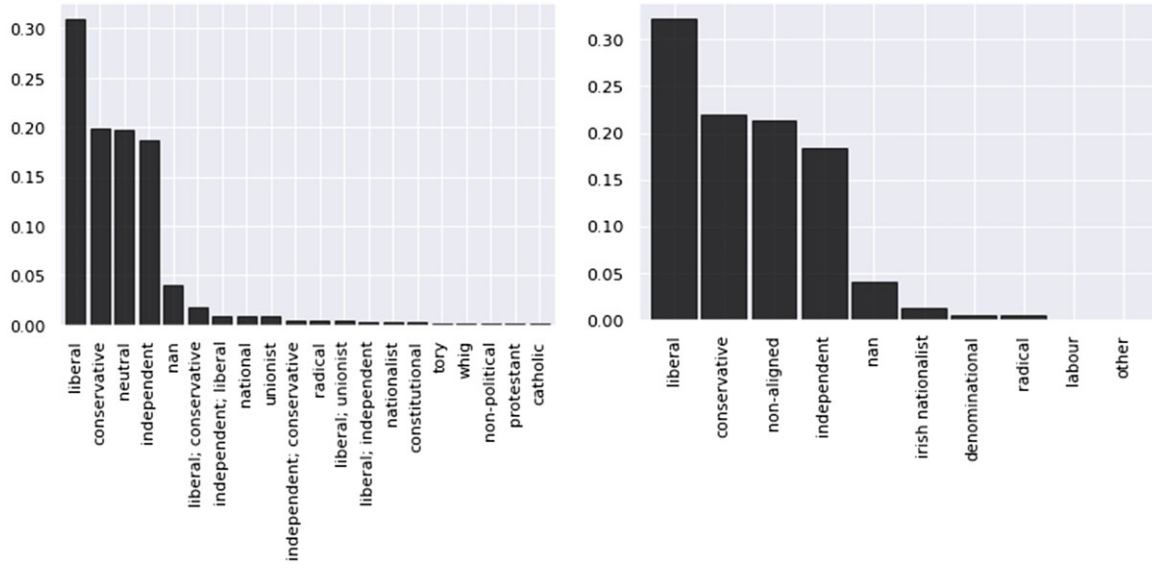


Fig. 5 Distribution of Political Labels. The left-hand side shows the top twenty original labels, the right hand side shows the simplified classification scheme. *X*-axis shows political labels, *Y*-axis indicates the proportion of each label.

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

corpus (marked by cross) and Mitchell (marked by a square).

For most of the nineteenth century, liberal newspapers took the largest share of provincial titles listed in the *Directories*. As the proportion of conservative periodicals declines during the third quarter of the century, the gap with the liberal titles widens. Both orientations converge again around the turn of the century, settling just above 20%. It is noticeable how the temporal patterns differ in the digital sample. Seen from the perspective of Victorian metadata, the JISC sample's curators clearly oversampled conservative newspapers: the sample's ratio of conservative titles is higher compared to the population. The conservative press is over-represented at the outset, accounting for almost 60% of all titles in the JISC corpus, but only 40% in Mitchell, while these percentages change, the relative gap between sample and population remains stable. The discrepancy between sample and population with regard to liberal papers increases slightly over time: the share of liberal titles remains stable at around 40%, and the sample reflects the population fairly well.

JISC leans conservative; nonetheless, it manages broadly to balance the prominent political perspectives of the day. Which raises the question: what type of representativeness is desirable? There exists, here, an unresolved tension between the wish to achieve equality—i.e. grant different voices equal presence—and the goal of proportionality: the oversampling among conservative titles places them on an equal footing with liberal papers, but this could also have unintended effects, such as marginalizing voices outside the liberal/conservative partisan dichotomy.

So far, we have established that in terms of proportionality, the JISC corpus is not representative, but can we, nonetheless, describe it as a balanced corpus? The attention to liberal and conservative papers comes at a cost and a more problematic picture appears when looking at 'non-aligned' and 'independent' papers. Neutral papers declined over the Victorian period, but the inverse holds for independent news, which grew in popularity. Taken together these labels constitute the majority of papers for the largest part of the Victorian era. From the 1860s onwards, around 40% of titles are either 'neutral' or 'independent', but these remain substantially underrepresented among JISC titles at around 10% of the overall corpus. While the digital sample aimed to balance liberal and conservative perspectives, it significantly downplayed the presence of neutral and independent perspectives.

While Fig. 6 reveals the principal evolutions over time, combining these trends in an overall score (or weighing their individual impact on the corpus's imbalance) remains difficult. We, therefore, analyse bias more rigorously using measures taken from information theory. We quantify the extent to which the sample deviates from the population using the

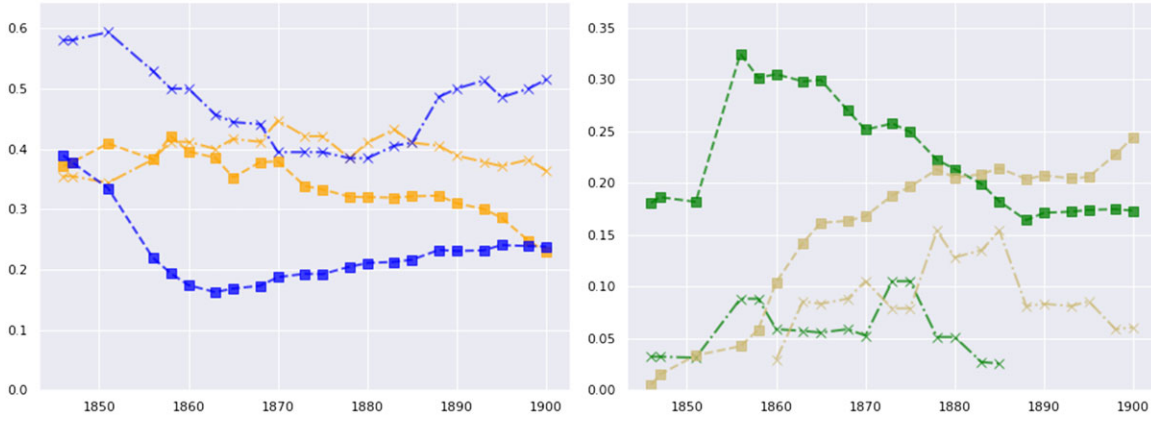


Fig. 6 (Left) Proportion (*Y*-axis) of 'liberal' (orange) and 'conservative' papers (blue) over time. (Right) Proportion of 'neutral' (green) and 'independent' (yellow) newspapers. Proportion for JISC newspapers is indicated with a cross marker, Mitchell with a square marker.

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

Jensen–Shannon Divergence (JSD), a common method for assessing the similarity between two probability distributions. We create two vectors $\overrightarrow{p}$ and $\overrightarrow{q}$, capturing the probability of political labels in both sample and population (we used the original as well the simplified label taxonomy). JSD is then expressed as the sum of two Kullback–Leibler (KL) divergences (Klingenstein *et al.*, 2014):

$$\mathrm{JSD}\left(\overrightarrow{p}, \overrightarrow{q}\right) = \frac{1}{2}\left[\mathrm{KL}\left(\overrightarrow{p} \middle| \frac{1}{2}\left(\overrightarrow{p} + \overrightarrow{q}\right)\right) + \mathrm{KL}\left(\overrightarrow{q} \middle| \frac{1}{2}\left(\overrightarrow{p} + \overrightarrow{q}\right)\right)\right]$$

JSD tends to increase when the political composition of the JISC sample diverges from the one observed in the *Directories*. We repeated this comparison for each year, thus obtaining a measure to monitor representativeness over time. To assess what labels explain the divergence (and in which direction), we compute their individual contributions. For each label i we calculate the partial KLs:

$$\mathrm{KL}_i\left(\overrightarrow{p}, \overrightarrow{q}\right) = p_i \mathrm{loglog} \frac{2p_i}{p_i + q_i}$$

While these trends are not identical, the two timelines point in similar directions. The bias scores peak for the first time around 1860, declining steadily in the next decade and starting to climb again from approximately 1880 onwards until the end of the century. Any explanation of the trends in Fig. 7 remains speculative, but we suggest that digitization may create a path dependency that affects the representativeness of corpora. The fact that digitization projects select papers at the title level, and prefer long-running periodicals, means that when the landscape changes rapidly, divergences can emerge. If we were to fit a regression line on the timelines in Fig. 7, it would reveal a general upward trend. However, the relationship with time is more complex. While the political composition of the JISC corpus resembles Mitchell's in the 1840s, the first rapid expansion of the press during the 1850s and 1860s is associated with an increase in divergence. The press keeps expanding in the following decades, but the bias seems to decline until the 1880s. In other words, the representativeness of the digital collections is sensitive to time in a nonlinear fashion, further complicating the analysis of such collections. For historians, engaged in diachronic analysis, it is especially important to appreciate how digital newspaper collections do not provide stable perspectives: their composition changes over time, just as the wider landscape. This, of course, does not make these collections less rich and useful, but divergences should be considered when discussing findings extracted from digital collections such as the JISC corpus.

To explain the divergence observed in Fig. 6, we can plot the contribution of each label over time (see Fig. 8). Positive scores indicate underrepresentation (sampling more among these papers would reduce bias), whereas negative values suggest this feature accounts for a disproportionately large number of titles. Figure 8 reveals the way in which JISC's digital reconstruction of the Victorian press is a politicized one which, in particular, downplays a substantial group of provincial newspapers which abstained from a clear political affiliation. The conservative press consistently ranks as the most overrepresented for the whole period, followed by liberal papers, which slowly but surely increase their divergence measure over time. Our most significant finding is the JISC sample's failure to capture the emergence of neutral and independent newspapers, which, taken together, constituted the largest share of provincial titles.

5.2 Price and place as dimensions of bias

Besides political orientation, the directories contain other information that allows us to compare the digital sample with the broader population. Like declared politics, price can offer clues about a newspaper's likely audience. Figure 9 shows how the distribution of newspapers' prices changed drastically in the Victorian period. The repeal of different 'taxes on knowledge' significantly reduced cost and spurred the proliferation of the press (Wiener, 2009). From the mid-1850s, newspapers became substantially cheaper, and by the end of the century more than 80% of provincial newspapers cost one penny or less.[15]

This dramatic decrease in prices is present but less pronounced in the digital collection, which seems to have oversampled among the more expensive titles, especially in the third quarter of the century, following trends in the newspaper landscape with a substantial lag. Figure 10 shows the divergence in distributions

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

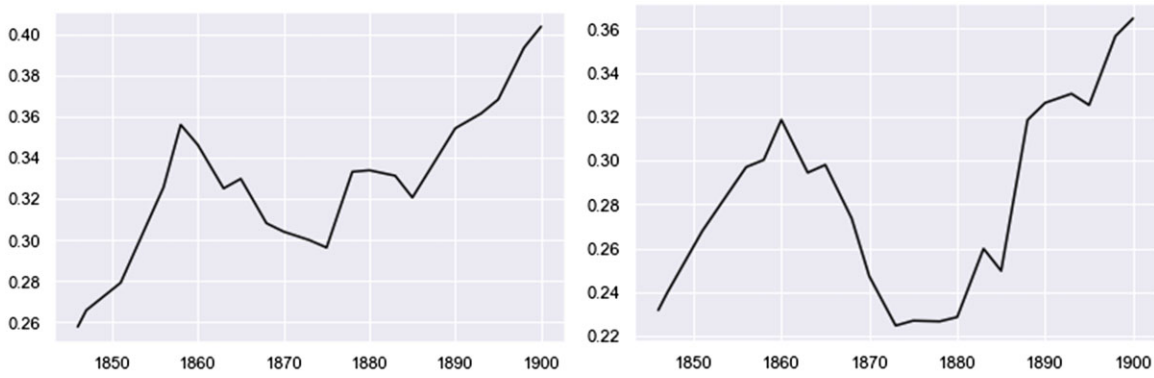


Fig. 7 Jensen–Shannon Divergence (*Y*-axis) between the probability distribution of political labels for JISC titles and Mitchell over time. The figure at the left-hand side is computed on the original labels, the one at the right-hand side on the clustered labels.

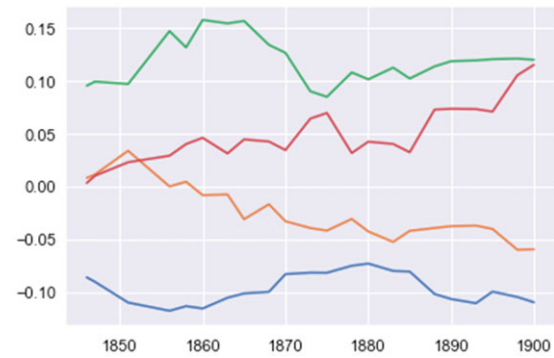


Fig. 8 Contribution of individual labels to the divergence. From top to bottom, the *Y*-axis shows the partial KL for neutral, independent, liberal, and conservative papers.

(left) and the contribution of titles at key price points (right), suggesting a grand arc over the second half of the nineteenth century: the divergence rises in the third quarter after which it generally declines.

As demonstrated by the timeline on the right-hand side of Fig. 10, the underrepresentation of cheaper prints drives most of the bias with respect to the price variable. Especially after the 1850s, the rapid rise of the one penny press caused a divergence between digital sample and population, the former containing the more 'expensive' papers that sold for 2d, 3d, or 4d. Penny and half-penny provincial titles pioneered the mass press in Britain, reaching a new audience of working- and lower-middle-class households living above the poverty line, but only just. By under-representing these titles, especially in the 1860s and 1870s, it seems likely that the JISC sample under-represents new, more demotic voices in the mid-Victorian British public sphere (Pooley, 2015).

Turning to place as a dimension of bias, here the results show less change over time as the divergence between sample and population (compared at the county level[16]) remains stable (see Online Appendix E). Partly this reflects the bias towards long-running titles in the JISC corpus, but it also suggests that the expansion of the provincial press was a largely even, nationwide phenomenon.[17] But did the JISC digitization programme succeed in its aim of offering a geographically balanced picture of the Victorian provincial press? To compute this, we calculated for each county the yearly contribution to the divergence, and averaged these scores.[18] This gives an indication of the overall under- or over-representation of a county in relation to the total number of titles published there.[19]

Figure 11 indicates how much each county contributes to the deviation between sample and population.[20] Overall, it shows an underrepresentation of the 'Home Counties', especially Kent, Sussex, Surrey, and Hertfordshire, and more attention to the industrial counties in the north of England (Lancashire, Yorkshire, and Northumberland, but not Cumberland). In Scotland, Aberdeenshire and Angus appear as oversampled, while the opposite holds true for Mid-Lothian.[21] There are also a group of counties in the English Midlands notably underrepresented: Lincolnshire, Staffordshire, and

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

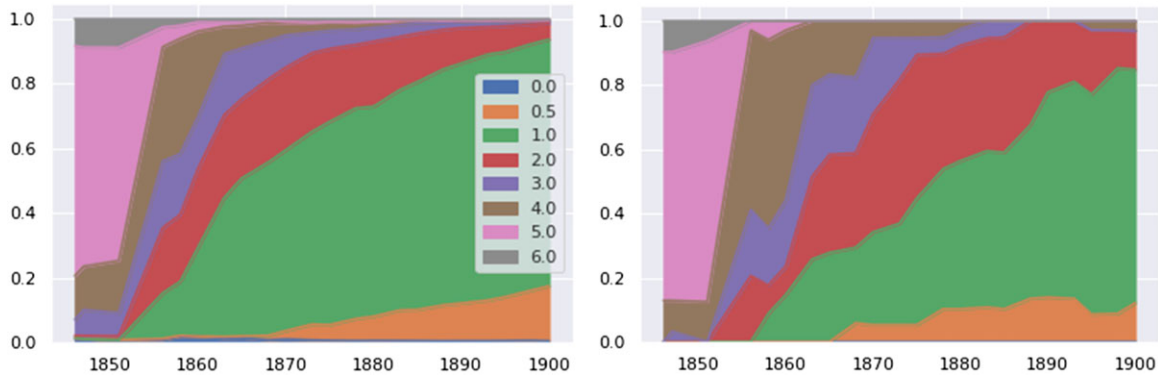


Fig. 9 Comparing the distribution of prices over time between the directories (left) and the JISC newspaper sample (right). *Y*-axis show cumulative proportion of each price point.

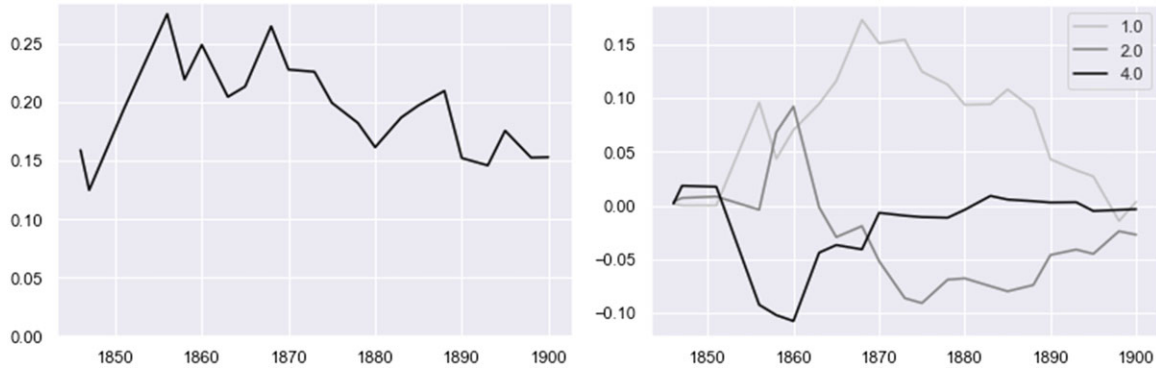


Fig. 10 The divergence in distributions (left) and contributions of papers sold at different prices (in pence; right). To make the figure more readable we only show the price points that most influenced the divergence over time (1d, 2d, and 4d).

to a lesser extent Shropshire, but there is no systematic regional bias here. In general, despite the under-representation of individual counties, the JISC sample appears to meet its aim to offer broad national coverage. The key exception—the region around London—may reflect an assumption that London dailies circulated more widely here, mitigating the need to digitize local titles.

6. Beyond Labels: Topics in Historical Descriptions of Newspapers

We can extend the Environmental Scan beyond the discrete labels and categories provided by Mitchell, by exploring the directories' original descriptions, or 'portraits' of newspapers. To do this, we followed a different procedure using Structural Topic Modelling (Roberts *et al.*, 2019), which estimates the association between topics and metadata variables. We estimated topic prevalence using YEAR (year in which the directory was published) and IN_JISC (a binary variable with value 1 if the paper is included in the JISC corpus, otherwise it is set to 0) as topic variables (otherwise known as independent variables). More specifically we used the following prevalence formula:

$$\text{Topic} \sim \text{YEAR} + \text{IN_JISC} + \text{YEAR} * \text{IN_JISC}$$

Figure 12 visualizes topic proportions over the whole corpus, showing the words with the highest

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

probability for each topic. Topic number 3, for example, appears as the largest, accounting for approximately 21% of the text in newspaper descriptions. It contains '*news*', '*local*', '*general*', '*give*', '*intellig*' as the most probable words. To establish the meaning of this (and other) topics, we further inspected word clouds (distant reading) and scrutinized text snippets that have a high probability for a topic (close reading).[22] Related to Topic 3, The *Woodford Times*, for example, is described as '[**giving**] the **local news** of, and a well-selected **digest** of **general intelligence**.' *The Echo*, in turn, 'gives all the **local** and district news, . . .'. Taken



Fig. 11 Partial KL scores averaged over all years by county, and then standardized over all counties. Scores are standard deviations from the mean with +3 (green) most overrepresented and −4 (red) most underrepresented. Colours correspond to percentiles, paler dots have scores closer to zero. See Footnote 19 to consult corresponding data directly

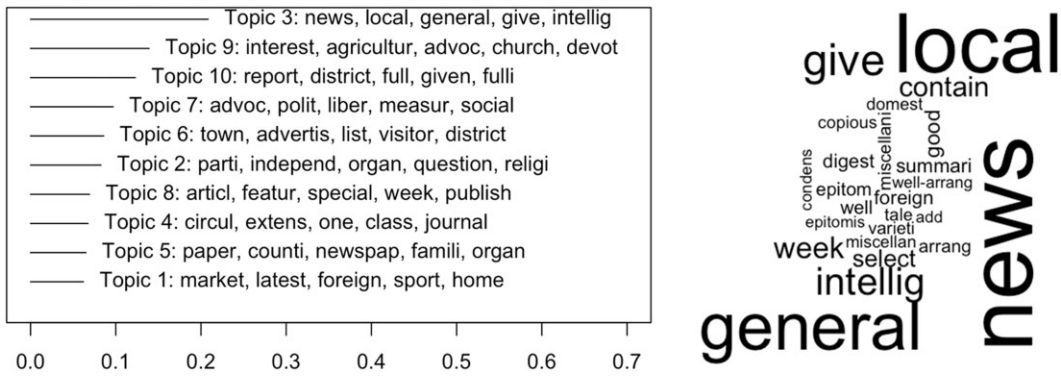


Fig. 12 Estimated topic proportions over all the newspaper descriptions (1846–1900) (left-hand side) and word cloud for Topic 3, the most prevalent topic (right-hand side)

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

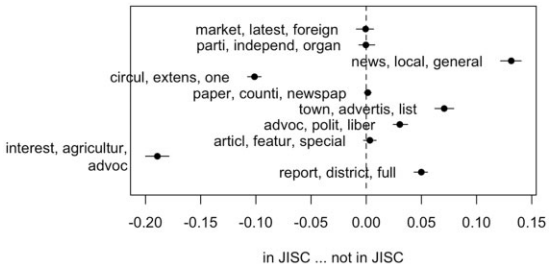


Fig. 13 Topic prevalence contrast for the IN_JISC variable (topics to the left of the axis line are associated with JISC titles, topics to the right are associated with the titles NOT in JISC)



Fig. 14 Word clouds for Topics 6 (left) and 10 (right) (Hobbs, 2018, 2013)

together, our analysis suggested that Topic 3 mainly captured 'local' newspapers, i.e. those emphasizing news for the locality or district.

To establish which type of newspaper is more likely to appear in the JISC corpus, we estimated the relationship between inclusion (IN_JISC) and the presence of each topic. Figure 13 shows prevalence contrast, estimating the association between topics and the inclusion or exclusion of newspaper titles from the JISC collection. Interestingly, the most prevalent topic in Mitchell—the one emphasizing the local orientation—seems the most underrepresented in the JISC corpus. As with the distribution of political labels, the Environmental Scan shows how a digital collection sometimes fails to capture substantial segments of the press that were contemporaneously very prominent. Like the 'neutral' titles, those emphasizing their 'local' focus are structurally underrepresented. This bias reflects historians' research priorities; political engagement over parochialism. By contrast, Victorian readers often valorized the local.

Similarly, Topics 6 and 10 are also underrepresented in the JISC corpus. Both clouds in Fig. 14 contain words that anchor newspapers in place: 'district', 'town', 'neighbourhood', '*local*' stressing the importance of locality for Victorian provincial newspapers and their audiences. Topic 6 is somewhat idiosyncratic in its local emphasis, primarily encompassing titles that functioned as directories, manuals to visitors and local advertising mediums.

Looking at other topics, the observed effects largely corroborate our earlier findings on political bias. While most topics lack an apparent political emphasis, ideological aspects were not completely absent either. There are two outliers—with respect to their political neutrality—Topic 7 ('*advoc*', '*polit*', '*liber*') and Topic 9 ('*interest*', '*agricultur*', '*advoc*'). Both the most probable words as well as the effect size exhibit a political orientation (results are not reproduced see Online Appendix D.1). Figure 15 arranges the most probable word stems in these topics as a cloud. The liberal orientation of Topic 7 is illustrated by entries such as '**Liberal** Non-conformist in politics, it **advocates social**, **religious**, and **political progress**' describing the *Bacup Times*. The slight under-representation of papers with an explicit liberal ethos (as captured by Topic 7), nuances the modest over-representation of Liberal labels as reported in Fig. 8. The largest effect in Fig. 13, however, is observed for Topic 9, the most 'conservative' leaning. The cloud visualization contains references to traditional tory priorities such as 'agriculture' and 'Church [of] England', but also signifies attention to socio-economic interests such as 'manufacturing' and 'commerce'. Mitchell's description of the *Cheltenham Journal* demonstrates how these disparate elements could be forged together. The paper 'advocates the interests of agriculture, commerce, and manufactures; is a political journal, devoting some space to literature and the fine arts, and attached to the Church of England.' Another aspect is politics, indicated by '*princip*' and '*poli*', and occasionally identified as conservative. The *Dorset County Chronicle* represents, in its own words, 'steadily pure Conservative principles'. The substantive effect towards IN_JISC, confirms our earlier finding on the over-representation of conservative papers. Lastly, Topic 4 also skews toward being over-represented.

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025



Fig. 15 Word clouds for the most liberal (left) and conservative (right) topics

The most probable word stems ('circul', 'extens', 'one') correspond to passages where Mitchell specifies circulation in social terms, for example, in phrases as '**circulates** amongst the upper and middle classes of society.' It is hard to interpret this result for understanding bias, but we speculate that the JISC curators' preference for newspapers of record brought with it a bias towards titles said to 'circulate' among socially select readerships.

7. Conclusion

In this paper, we introduce the Environmental Scan as a method for critically investigating the composition of digital collections. The scan reconstructs the landscape from which digital information is extracted. It articulates the relationship between digital sample and analogue whole. But, crucially, it does so by taking into account the historical 'vernacular', i.e. the original language and classifications that surrounded and shaped the source material. Using the example of the Victorian provincial press, we demonstrate how even large heritage collections can systematically under-represent important dimensions of the past; in this case, types of newspaper central to past taxonomies and social practices.

Analysing the JISC newspaper corpus using metadata derived from original Victorian reference works highlights significant biases in its composition. JISC's curators oversampled conservative provincial newspapers and, generally over-represented politically aligned newspapers at the expense of neutral and independent titles, thereby exaggerating the politicization of the Victorian public sphere and its obeisance to party politics. Similarly, our analysis revealed a preference for sampling more expensive newspapers, with more 'demotic' halfpenny and penny titles being under-represented. Looking at the topics that run through the descriptions of nineteenth-century newspapers, we observed a strong emphasis on the local and regional aspects of the press, which was not captured by the newspapers included in the JISC corpus.

Arguably, digital corpora reflect the priorities of twenty-first-century scholars over those of Victorian readers. But our aim is not to criticize sampling decisions, but rather to deepen a critical understanding of the data. Historical data will always contain biases. Although it can certainly be used to inform future digitization priorities, the point of the Environmental Scan is not to eliminate bias, but rather to provide scholars with tools that can help them understand and work with it. By better understanding the characteristics of a collection they will be better able to develop findings and interpretations sensitive to historical context. We offer the Environmental Scan as an innovative contribution that brings the traditions of source criticism to bear on the research materials of the digital age.

The methods and data introduced in this paper will be most directly valuable to researchers working with British newspaper collections. Contextual resources such as Mitchell's—and the 'Victorian metadata' derived from them—will allow scholars to refine their analysis of digital newspapers (for example by comparing sub-classes of newspapers by place, politics, and price). Here the focus has been on developing metadata derived from what historians would term 'actors' categories', but future work will delve deeper into the content of digital newspapers, for example, by exploring the relationship between newspaper partisanship and distinct patterns of language use.

The principles of the Environmental Scan have relevance beyond the specific case of British newspapers. Firstly, there is scope to develop comparable scans for other newspaper collections, especially since, across the world, considerable effort is being directed at compiling large, open-access collections of historical newspapers. Secondly, although Mitchell's press directories undoubtedly represent a particularly rich source of contemporaneous metadata, they are not

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

unique. For example, library catalogues, or short-title lists, can enable researchers to compute the relation between a digital sample of books and the broader universe of publications. Reconstructing historical classification schemes and labelling practices may be more challenging, but again not impossible. In the absence of a source such as Mitchell's scholars may be able to use 'vintage' classification labels such as those preserved by the French National Library or reconstruct past understandings from diverse contemporaneous sources, as Ted Underwood has done in his work on shifting conceptions of literary genres (Underwood, 2020).

In short, we see the ethos behind the Environmental Scan as being of broader significance for the future of digital scholarship. Its central contention is that, when addressing questions of bias in digital heritage, our goal should not be to use our insights to aim at curating a 'perfect' sample. This would be a fool's mission. Rather, it should be to use our enhanced understanding of the data to inform our analyses and nuance our interpretations. If 'Big Data' wants to convince sceptical scholars in the humanities then it needs to embrace methods that foreground contemporary meaning and context. This is the rationale for the Environmental Scan.

Funding

Work for this paper was produced as part of Living with Machines. This project, funded by the UK Research and Innovation (UKRI) Strategic Priority Fund, is a multidisciplinary collaboration delivered by the Arts and Humanities Research Council (AHRC grant AH/S01179X/1), with The Alan Turing Institute, the British Library and the Universities of Cambridge, East Anglia, Exeter, and Queen Mary University of London. This work was also supported by The Alan Turing Institute (EPSRC grant EP/N510129/1).

Supplementary data

Supplementary data are available at *DSH* online.

References

Algee-Hewitt, M., Allison, S., Gemma, M., Heuser, R., Moretti, F., Walser, H. (2016). Canon/archive. Large-scale dynamics in the literary field, Literary Lab Pamphlet 11. https://litlab.stanford.edu/LiteraryLabPamphlet11.pdf.

Beales, D. (1967). Parliamentary parties and the "independent" member, 1810-60. In Robson, R. (ed.), *Ideas and Institutions of Victorian Britain: Essays in Honour of George Kitson Clark*. London: G. Bell, pp. 1–19.

Biber, D. (1993). Representativeness in corpus design. *Literary and Linguistic Computing*, **8**(4): 243–57.

Bode, K. (2017). Fictional systems: Mass-digitization, network analysis, and nineteenth-century Australian newspapers. *Victorian Period Review*, **50**(1): 100–38. https://doi.org/10.1353/vpr.2017.0005

Bode, K. (2020). Why you can't model away bias. *Modern Language Quarterly*, **81**: 95–124. https://doi.org/10.1215/00267929-7933102

Brake, L. (2015). Nineteenth-century newspaper press directories: the national gallery of the British Press. *Victorian Period Review*, **48**: 569–90. https://doi.org/10.1353/vpr.2015.0055

Brake, L. (2012). Half full *and* half empty. *Journal of Victorian Culture*, **17**: 222–29. https://doi.org/10.1080/13555502.2012.683149

Chasalow, K. and Levy, K. (2021). Representativeness in statistics, politics, and machine learning. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*. Presented at the FAccT '21: 2021 ACM Conference on Fairness, Accountability, and Transparency, ACM, Virtual Event Canada, pp. 77–89. https://doi.org/10.1145/3442188.3445872

Cordell, R. (2017). "Q i-jtb the Raven": taking dirty OCR seriously. *Book History*, **20**: 188–225. https://doi.org/10.1353/bh.2017.0006

Fyfe, P. (2018). Access, computational analysis, and fair use in the digitized nineteenth-century press. *Victorian Period Review*, **51**: 716–37. https://doi.org/10.1353/vpr.2018.0051

Fyfe, P. (2016). An archaeology of Victorian newspapers. *Victorian Period Review*, **49**: 546–77. https://doi.org/10.1353/vpr.2016.0039

Garg, N., Schiebinger, L., Jurafsky, D., and Zou, J. (2018). Word embeddings quantify 100 years of gender and ethnic stereotypes. *Proceedings of the National Academy of Sciences of the United States of America*, **115**: E3635–44. https://doi.org/10.1073/pnas.1720347115

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

Gliserman, S. (1969). Mitchell's "Newspaper Press Directory": 1846–1907. *Victorian Periodicals Newsletter*, **4**(2): 10–29.

Hauswedell, T., Nyhan, J., Beals, M.H., Terras, M., and Bell, E. (2020). Of global reach yet of situated contexts: an examination of the implicit and explicit selection criteria that shape digital archives of historical newspapers. *Archaelogical Science*, **20**: 139–65. https://doi.org/10.1007/s10502-020-09332-1

Hills, T.T., Proto, E., Sgroi, D., and Seresinhe, C.I. (2019). Historical analysis of national subjective wellbeing using millions of digitized books. *Nature Human Behaviour*, **3**: 1271–5. https://doi.org/10.1038/s41562-019-0750-z

Hobbs, A. (2018). *A Fleet Street in Every Town: The Provincial Press in England, 1855-1900*. Cambridge, UK: Open Book Publishers. https://doi.org/10.11647/OBP.0152

Hobbs, A. (2013). The deleterious dominance of *The Times* in nineteenth-century scholarship. *Journal of Victorial Culture*, **18**: 472–97. https://doi.org/10.1080/13555502.2013.854519

Holahan, C. (2021). Rummaging in the dark: ECCO as opaque digital archive. *Eighteenth-Century Studies*, **54**: 803–26. https://doi.org/10.1353/ecs.2021.0093

King, E. (2008). Digital historic newspapers online: prospects and challenges. In Walravens, H. (ed.), Newspapers Collection Management: Printed and Digital Challenges: Proceedings of the International Newspaper Conference, Santiago de Chile, 3–5 April 2007. K. G. Saur, pp. 59–78. https://doi.org/10.1515/9783598441011.59

King, E. (2005). Digitisation of newspapers at the British Library. *Serials Librarian*, **49**: 165–81. https://doi.org/10.1300/J123v49n01_07

Klingenstein, S., Hitchcock, T. and DeDeo, S. (2014). The civilizing process in London's Old Bailey. *Proceedings of the National Academy of Sciences of the United States of America*, **111**: 9419–24. https://doi.org/10.1073/pnas.1405984111

Lahti, L., Mäkelä, E., Tolonen, M. (2020). Quantifying Bias and Uncertainty in Historical Data Collections with Probabilistic Programming. In Karsdorp, F., McGillivray, B., Nerghes, A. and Wevers, M. (eds). *CHR2020 Computational Humanities Research 2020: Proceedings of the Workshop on Computational Humanities Research*, Amsterdam, the Netherlands, November 18–20, 2020.

Lansdall-Welfare, T., Sudhahar, S., Thompson, J., Lewis, J., FindMyPast Newspaper Team, and Cristianini, N. (2017). Content analysis of 150 years of British periodicals. *Proceedings of the National Academy of Sciences of the United States of America*, **114**: E457–E465. https://doi.org/10.1073/pnas.1606380114

Lee, A. (1976). The origins of the Popular Press, 1855-1914. London: Croom Helm.

Mak, B. (2014). Archaeology of a digitization. *Journal of the Association for Information Science and Technology*, **65**(1): 1515–26. https://doi.org/10.1002/asi.23061

Michel, J.-B., Shen, Y.K., Aiden, A.P. et al. (2011). Quantitative analysis of culture using millions of digitized books. *Science*, **331**: 176–82.

Mussell, J. (2012). Nineteenth-century newspapers in the digital age. *Journal of Victorian Culture*, **17**: 221. https://doi.org/10.1080/13555502.2012.683147

O'Malley, T. (2015). Mitchell's Newspaper Press Directory and the Late Victorian and Early Twentieth-Century Press. *Victorian Period Review*, **48**: 591–606. https://doi.org/10.1353/vpr.2015.0057

Pechenick, E.A., Danforth, C.M., and Dodds, P.S. (2015). Characterizing the Google Books Corpus: strong limits to inferences of socio-cultural and linguistic evolution. *PLoS One*, **10**: e0137041. https://doi.org/10.1371/journal.pone.0137041

Pitkin, H.F. (1967). *The Concept of Representation*. Berkeley: University of California Press.

Pooley, S. (2015). Children's writing and the popular press in England, 1876-1914. *History Workshop Journal*, **80**(1): 75–98.

Poovey, M. (1995). *Making a Social Body: British Cultural Formation, 1830-1864*. Chicago, IL: Chicago University Press.

Roberts, M.E., Stewart, B.M. and Tingley, D. (2019). stm: an *R* package for structural topic models. *Journal of Statistical Software*, **91**: 1–40. https://doi.org/10.18637/jss.v091.i02

Shaw, J. (2009). British newspapers 1620-1900: Final Report, in: JISC and the British Library, August 2009. https://www.webarchive.org.uk/wayback/archive/20140613220103/http://www.jisc.ac.uk/media/documents/programmes/digitisation/blfinal.pdf.

Shaw, J. (2006). 10 billion words: the British Library British Newspapers 1800-1900 Project: some guidelines for large-scale newspaper digitisation. *IFLA PUBLICATIONS*, **118**: 27.

Underwood, T. (2020). Machine learning and human perspective. *PMLA*, **135**(1): 92–109.

Vincent, D. (2014). The invention of counting: the statistical measurement of literacy in nineteenth-century

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

England. *Comparative Education*, **50**(3): 266–81. https://doi.org/10.1080/03050068.2014.921372.]

Wiener, J. (2009). Newspaper Taxes, Taxes on Knowledge, Stamp Taxes. In Brake, L. and Demoor, M. (eds), *The Dictionary of Nineteenth-Century Journalism in Great Britain and Ireland*. London: Academic Press and the British Library, pp. 1265–66.

Wiener, J. (1969). *The War of the Unstamped: the Movement to Repeal the British Newspaper Tax*, 1830–1836. Ithaca, NY: Cornell University Press.

Yale, E. (2015). The history of archives: the state of the discipline. *Book History*, **18**: 332–59. https://doi.org/10.1353/bh.2015.0007

Notes

1 For example, the twentieth century saw an increase in scientific texts, which called into question many of the findings extracted from the Google Books data.

2 Headline and quotation are taken from *The Times*, 15 October 2019. The online version of the article (with a different title) can be accessed here: https://www.thetimes.co.uk/article/victorian-times-were-happiest-study-of-the-national-mood-finds-5sqdl5cmh

3 For full details of the collections and current re-use arrangements see https://www.digitisednewspapers.net/histories/bl/ (accessed 12 May 2022).

4 Open access initiatives such as the British Library's 'Heritage Made Digital' (HMD) are dwarfed by the ongoing commercial digitization partnership between the British Library and FindMyPast (FMP). Whereas the selection criteria used by HMD are made public, those used by FMP are unclear and probably reflect the needs of genealogists rather than researchers.

5 Published quarterly from 1867 and indexed retrospectively back to 1790, see http://c19index.chadwyck.com/marketing/aboutpalmers.jsp

6 https://www.britishnewspaperarchive.co.uk/ (accessed 17 September 2021).

7 For a list of the JISC titles included in this study see the GitHub repository: https://github.com/Living-with-machines/PressDirectories. The table 'JISC_TitleList.csv' shows all the titles included in the JISC corpus. In addition, the table 'jisc_links.csv' shows how newspaper titles were linked to the press directories.

8 The higher price included the penny stamp for delivery by mail.

9 The data and digitization workflow are described in this GitHub repository: https://github.com/Living-with-machines/PressDirectories

10 Years for which we manually corrected the data are 1846, 1847, 1851, 1856, 1858, 1860, 1863, 1865, 1868, 1870, 1873, 1875, 1878, 1880, 1883, 1885, 1888, 1890, 1893, 1895, 1898, 1900 and 1905, 1910, 1915, 1920 (the latter years are not part of this paper).

11 The British Library provided us with a spreadsheet that records the newspaper titles included in the JISC collection. Table is available in the GitHub repo https://github.com/Living-with-machines/PressDirectories). We thank Yann Ryan and Luke McKernan for sharing this information.

12 For 1 year, 1873, the orange line overtook the blue line, this is because one title was ambiguous and we could not decide which entry in Mitchell it corresponded with. Appendix B also lists the titles which we failed to link with Mitchell's *Directories* in specific years.

13 We distinguished the following head labels, discarding moderators (like 'ultra' or 'moderate', etc.): 'anti-whig', 'catholic', 'church of england', 'church of ireland', 'conservative', 'constitutional', 'evangelical', 'free-trade', 'independent', 'labour', 'liberal', 'national', 'nationalist', 'neutral', 'no-politics', 'non-comformist', 'non-party', 'non-political', 'non-sectarian', 'orange', 'presbyterian', 'progressive', 'protestant', 'radical', 'repeal', 'republican', 'tory', 'ultramontane', 'unionist', 'whig'.

14 The simplified schema (and rationale behind reclassifications) can be found in Online Appendix C. It contains labels: 'liberal', 'conservative', 'non-aligned', 'independent', 'irish nationalist', 'denominational', 'radical', 'labour', and 'other'. We wish to point out here that, given the importance of classifications in this paper, decisions were discussed intensively among the historical experts on the Living with Machine project. Moreover, we've repeated all the results below with the original labels, to ensure that our choices in this respect did not skew our results or interpretation.

15 To facilitate visualization of trends over time (in Figs 11 and 12), we processed the prices as follows: if two prices are mentioned, we only take the first, then we mapped all price mentions (represented as strings) to a number, expressing the price in number of pennies, i.e. '2 ½ d' was mapped to 2.5, 'Gratis' to 0 and '4s' to 48. Records for which we could not find a price point were discarded, however, including them and setting their values to 0 or 6 (the maximum value) does not change the results as only 1,143 records (2.6%) lacked this information. We simplified prices further by setting the maximum price to a sixpence (a fortune for the Victorian

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025

reader, and newspapers rarely cost more). Lastly, after we converted all prices to decimal numbers, and rounded upwards to the nearest integer (i.e. 3.5 become 4) except the halfpenny papers, which appeared as a prominent and distinct category of papers catering to a wide range of often less affluent audiences.

16 We compute the distribution of papers by county as a proxy for geographical spread. The JSD scores remain rather stable over the whole period.

17 One issue with JSD scores is that they are difficult to compare to other settings and therefore to establish if the divergence is high or higher than (e.g. compared to those observed when analysing political labels).

18 More precisely we mapped counties recorded in Mitchells to the ids of historic counties in Wikidata. We used these ids (not county names) to compute the divergence.

19 Please note that we do not compare the sample to the number of residents but to the number of newspapers published in each county.

20 The original map with all scores can be accessed here: https://kepler.gl/demo/map?mapUrl=https://dl.dropboxusercontent.com/s/s7tv62okhoatt1u/keplergl_dhmlpoi.json (accessed 18 October 2021).

21 We mapped all county names in Mitchell to (the id of) historic counties in wikidata. The county names in Fig 12 are the names associated with the wikidata ids (Mitchell used 'Edinburghshire' for 'Midlothian').

22 We selected text snippets that have a high probability for a given topic. Moreover, we also applied other weighting schemes to generate word profiles. Besides only showing the most probable words of a topic, we used other measures to highlight more distinctive words. For example, FREX weights words by their overall frequency and exclusivity for the topic at hand. For Topic 3, these include 'select', 'summari', 'epitom'.

Downloaded from https://academic.oup.com/dsh/article/38/1/1/6644524 by guest on 26 June 2025